TAO retail product detection and recognition

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> Applicable environment: Jetson Orin NX, JetPack 7.2.1, DeepStream 9.1.

1. Case goal and processing flow

Retail Object Detection (DINO) is responsible for locating the products on the shelf. It is not a single-label classification of the whole shelf image.

```
Video/USB/CSI → the product detection (Retail DINO) → the preview and the video output
```

2. The checks before running it

```
cd ~/deepstream_tao_apps/deepstream_app_tao_configs

# Confirm the configuration and the input source
grep -E "uri=|config-file=" run_retail.txt

# Confirm the custom ONNX and the engine
ls -la ~/deepstream_tao_apps/models/retail/

# Confirm the original model (it comes with the DS)
ls -la /opt/nvidia/deepstream/deepstream-
9.1/samples/models/tao_pretrained/retail_object_detection_vdeployable_100_onnx_v1.0/
```

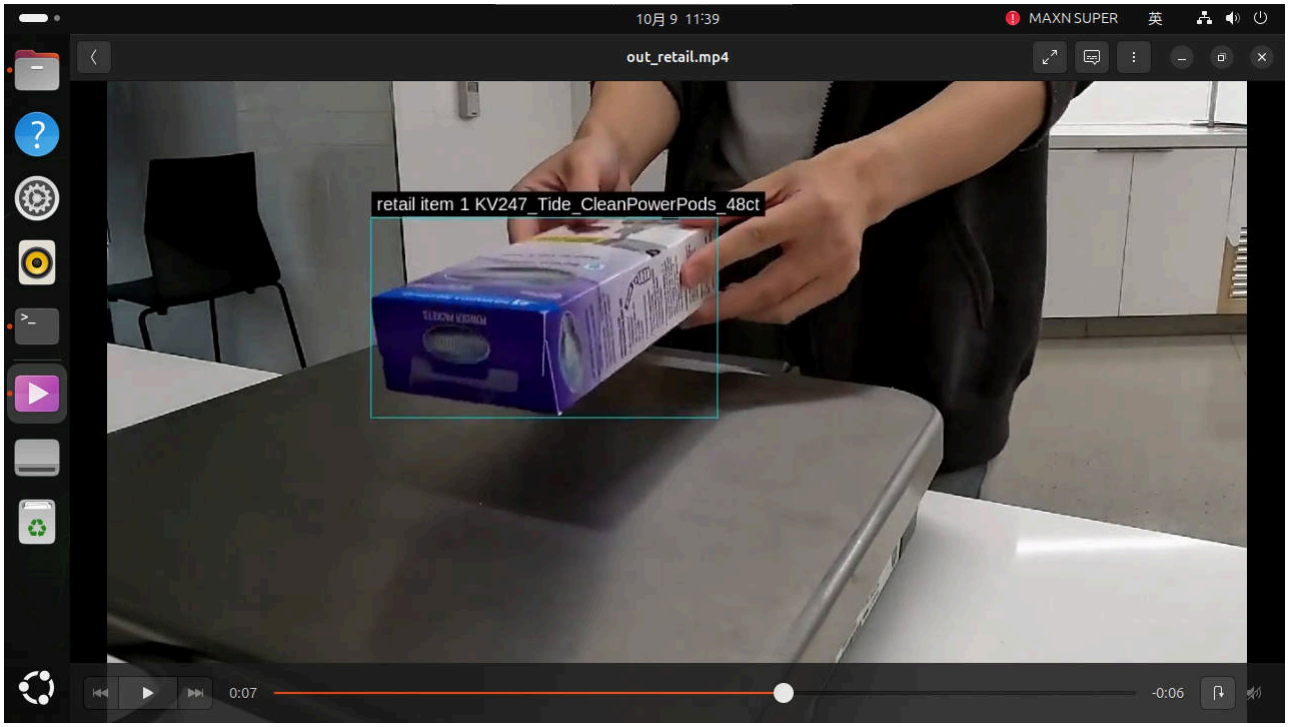
Confirm that the material and the output directory are available. The camera mode needs the graphical desktop/VNC; for the USB check the device node and for the CSI confirm that the sensor and the Argus are ready.

Key: the `score_threshold` of the Retail DINO ONNX has been changed from 0.4 to **0.1**, and it is stored as `~/deepstream_tao_apps/models/retail/retail_detector_100_v2.onnx`.

3. Start the detection

The video mode:

```
export DISPLAY=:0 && ~/deepstream_tao_demo/run_case.sh retail video
```



The input is the retail video `sample/streams/retail_object_h264.mp4` that comes with DeepStream. The detection boxes and the category labels are the product of the same stage; the course configuration uses the cyan-blue detection boxes that are consistent with the tutorial.

After the EOS, check the output of the pipeline. The current configuration generates `retail_out.mp4`:

```
ls -la ~/deepstream_tao_apps/deepstream_app_tao_configs/retail_out.mp4
```

To replace the test video: modify the `uri=file://` of the `[source0]` section in `run_retail.txt` to the absolute path of a retail MP4/H.264 file on the Jetson itself. Use `gst-discoverer-1.0` to confirm that it can be decoded; it is best if the material contains clear products on the shelf. Keep the original video and the outputs that you need.

The first run compiles the Retail DINO engine (about 11 minutes); the second start loads it instantly.

4. The real-time detection with a USB camera

Connect the camera, check the node and the formats and then start it in a graphical session:

```
ls -l /dev/video*
v4l2-ctl -d /dev/video0 --list-formats-ext

export DISPLAY=:0 && ~/deepstream_tao_demo/run_case.sh retail usb
```



The script detects the USB camera node automatically through `~/deepstream_tao_demo/find_usb_cam.sh`.

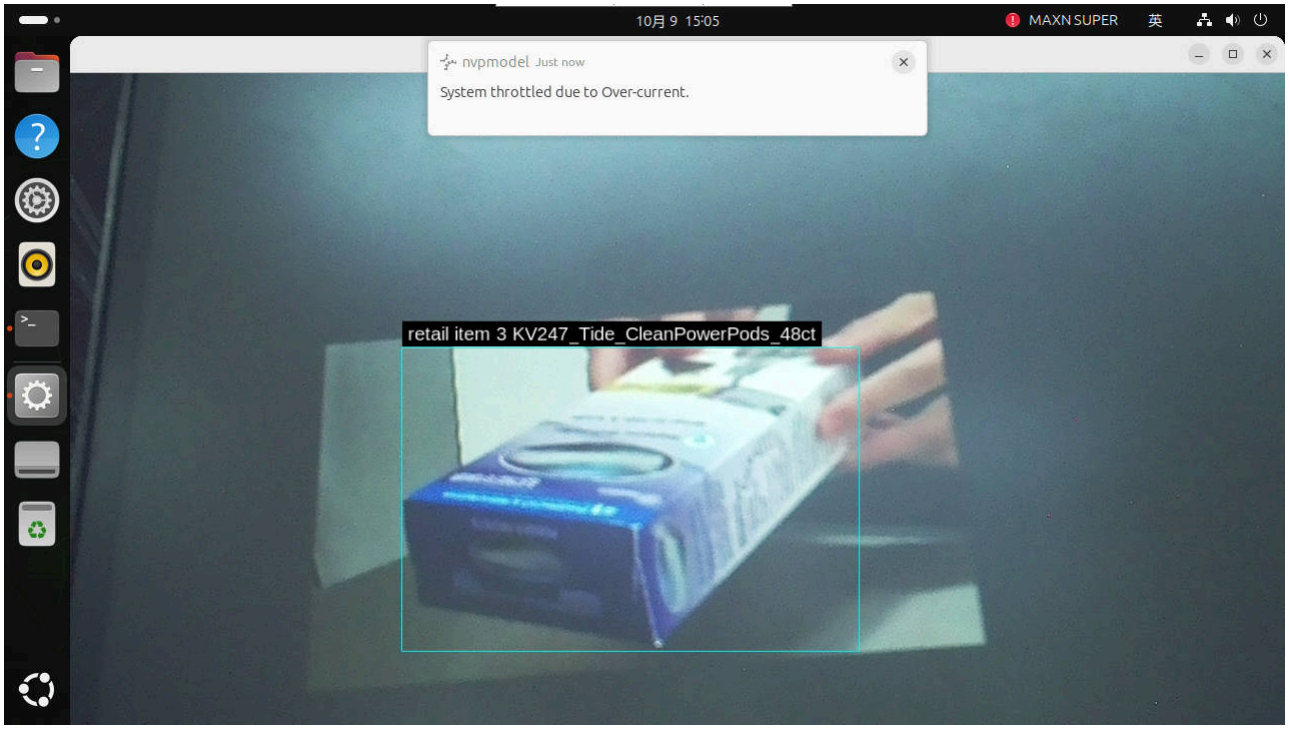
The real-time camera entry only previews and it does not produce an offline MP4.

Press `Ctrl+C` once, wait for the application to exit and confirm that the camera has been released. The Retail DINO has a high load; on the Orin NX the video mode is about 21 FPS and the USB mode is about 15 FPS.

5. The real-time detection with a CSI camera

Start:

```
export DISPLAY=:0 && ~/deepstream_tao_demo/run_case.sh retail csi
```



The CSI camera entry supports the real-time preview and it does not produce an offline video output. To stop it, press `Ctrl+C` once and wait for the application and the camera streaming to exit.

6. Frequently asked questions

Symptom	The check and the handling
The products cannot be detected	Look at the log of the detection model, at the scale of the image and at the occlusion; confirm that the input material contains a shelf.
The labels are garbled (an INT32 parsing error)	Confirm that the `pgie` configuration uses `NvDsInferParseCustomEfficientDetTAO` (and not the default `NvDsInferParseCustomBatchedNMSTLT`).
There are too many boxes/the boxes are noisy	Confirm that the `score_threshold` of the ONNX has been changed from 0.4 to 0.1; `cluster-mode=2` + `nms-iou-threshold=0.3`.
`Failed to load config file`	Check the `config-file=` path in `run_retail_csi.txt` (the trap of the path missing the `configs/` prefix was hit before).
The engine fails to load	The old engine was built with TRT 10 and it fails under DS 9.1; delete `~/deepstream_tao_apps/models/retail/*.engine` so that the DS rebuilds it automatically.
There is no image with USB	Check the node, the format, the permissions, the occupancy and the display session.
There is no image with CSI	Check the connection of the compatible sensor, the driver, the Argus service and whether the camera is in use; this entry directly uses the Argus `type=5`.
The accuracy of the labels is limited	The model only recognizes the 100 product categories of the training set; a product that is not in the training set gets the "most similar" category.
`nvv4l2h264enc has no property named 'gpu-id'`	It is a known non-blocking warning of DS 9.1 and it does not affect the inference or the display.

7. References

- The NVIDIA DeepStream TAO application examples: https://github.com/NVIDIA/DeepStream/tree/main/src/apps/tao_apps
- The DeepStream TAO integration documentation: https://docs.nvidia.com/metropolis/deepstream/dev-guide/text/DS_TAO_integration.html